

1(a)(i)	<i>all correct for one mark ;</i> <table><tr><td>palmitic acid</td><td>0</td></tr><tr><td>linoleic acid</td><td>2</td></tr><tr><td>stearic acid</td><td>0</td></tr><tr><td>palmitoleic acid</td><td>1</td></tr></table>	palmitic acid	0	linoleic acid	2	stearic acid	0	palmitoleic acid	1	1
palmitic acid	0									
linoleic acid	2									
stearic acid	0									
palmitoleic acid	1									
1(a)(ii)	circle drawn around palmitic acid <u>and</u> around stearic acid ;	1								
1(a)(iii)	COOH / carboxyl (group) / carboxylic acid (group) ; A carbonyl group and hydroxyl group R if response includes incorrect groups	1								

Question	Answer	Marks
1(a)(iv)	<i>if no ref. to fatty acids, allow one mark for (some) triglycerides may, be saturated / have no double bonds (in hydrocarbon chain,) <u>and</u> (some) may, be unsaturated / have double bonds</i> I 'tails' any two from: - each triglyceride has three fatty acids ; - fatty acids can be, <u>saturated</u> / with no double bonds, <u>and</u>, <u>unsaturated</u> / with double bonds ; A unsaturated fatty acids can have a different number of double bonds - fatty acids can have different, (hydrocarbon chain) lengths / number of carbons ; - detail from Table 1.1 to support mp2 or 3 ; saturated = palmitic / stearic v unsaturated = oleic / linoleic / palmitoleic or 1 double bond = oleic / palmitoleic v 2 double bonds = linoleic or oleic / linoleic / stearic = 18C v palmitic / palmitoleic = 16C <i>idea of</i> a particular fatty acid, can be located at a different position in a triglyceride / can form a bond with any one of the three (functional) hydroxyl groups ; <i>ref. to</i> different environmental conditions of plants grown in different regions ; e.g. colder conditions and oils having (more triglycerides with) higher proportion of unsaturated fatty acids <i>idea of</i> (shared triglycerides) have structures (most) suited as energy stores ; A example e.g. (have fatty acids with) longest hydrocarbon chains easiest to, hydrolyse / release hydrogen atoms have greater energy value than others <i>ref. to</i> ease of, production / synthesis ; <i>idea that</i> shared triglycerides composed of the fatty acids (present) in highest proportions ; e.g. <i>using Table 1.1 to determine the most common 2 or 3 fatty acids</i> 'could all have the same fatty acid combination from oleic acid and / or palmitic acid and/or linoleic acid' (any two)	2

Question	Answer	Marks
1(b)	any two from: 1 water is, a polar molecule / polar / dipolar ; A water is a dipole A description e.g. the O atom (of water) has, a slightly negative / δ^-, charge, <u>and</u> the H atoms have a slightly positive / δ^+, charge 2 water can form, hydrogen bonds / H-bonds, with, glycerol / polar substances ; ora = water cannot form H-bonds with, triglycerides / non-polar substances 3 glycerol / polar molecules / ions, <u>interact</u> with water (molecules) / are attracted to water (molecules) / are hydrophilic ; A idea that water, attracts / collects around / AW, ions / charged substances ora = triglycerides / non-polar / uncharged, molecules, do not interact with water / repel water / are hydrophobic I 'tails'	2
1(c)	(phloem) <u>sieve tube element</u> ; (<i>question asks for type of cell</i>) A sieve element	1